UMass Boston

PEAAII AI Applications Hackathon 2026
Agentic AI, October 2026

Roameo

plans your trip and lets you decide on the money

Try it
roameo-l13h.onrender.com

AI agents do the legwork.

Σ Code checks the money.

You make the call.

Code
github.com/TinyZhen/Travel_Agent

An agent here is an AI helper with one job.

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Alex has $600 for four days in Chicago and types one sentence



Alex

a student we made up

$600, 4 days, Chicago

What Alex types in Roameo

Chicago for 4 days, budget $600

What has to add up, and again after every change

flights + hotel + sights + food + transit

= ? within the **$600**, or not?

A chat reply can do the sum itself and say it fits your budget, with nobody checking it.

89% of travelers want AI help to plan a trip, but only 12% are happy to let AI decide alone [2].

[2] Booking.com, Global AI Sentiment Report, July 2025. Alex is an illustration. The run on the next slides is real.

ROAMEO RESTS ON THREE SIMPLE IDEAS

AI agents do the legwork

what the AI decides

a They ask for what is missing

b They choose which searches the trip needs

c They turn your plain words into changes

IN ALEX'S RUN

Asked Alex where from and when, then compared 8 flights, 18 hotels and 59 events

Code checks the money

what plain code decides

a It adds up every total

b It tries one fix at a time, at most five

c It shows the gap and never hides it

IN ALEX'S RUN

Took the plan from $1,235 to $780, still $180 over Alex's $600

You make the call

what you decide

a You choose how to close the gap

b You see every step and every assumption

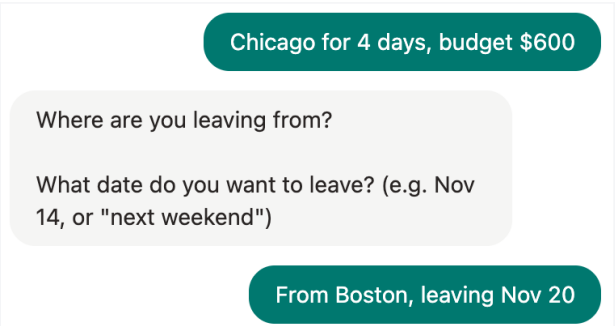
c Nothing is final until you say OK, and you book it yourself

IN ALEX'S RUN

Asked Alex whether to raise the budget, stay 2 nights or drive

AI AGENTS ASK FIRST, THEN CHOOSE WHAT TO SEARCH

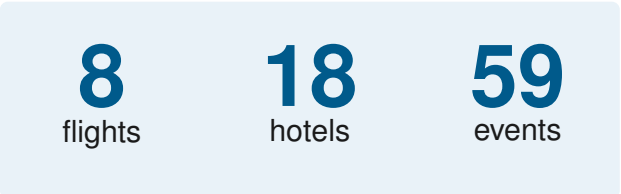
a They ask for what is missing



It never guesses where you leave from or when, because a guessed city or date gives a useless plan.

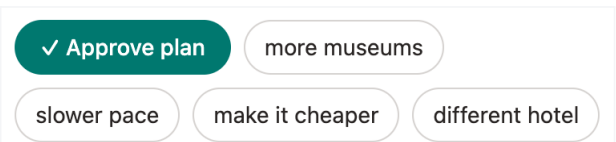
Alex's live run. An agent here is an AI helper with one job.

b They choose which searches the trip needs



Past years' weather said cold and rainy, so it also looked for indoor places, holiday markets and ice skating.

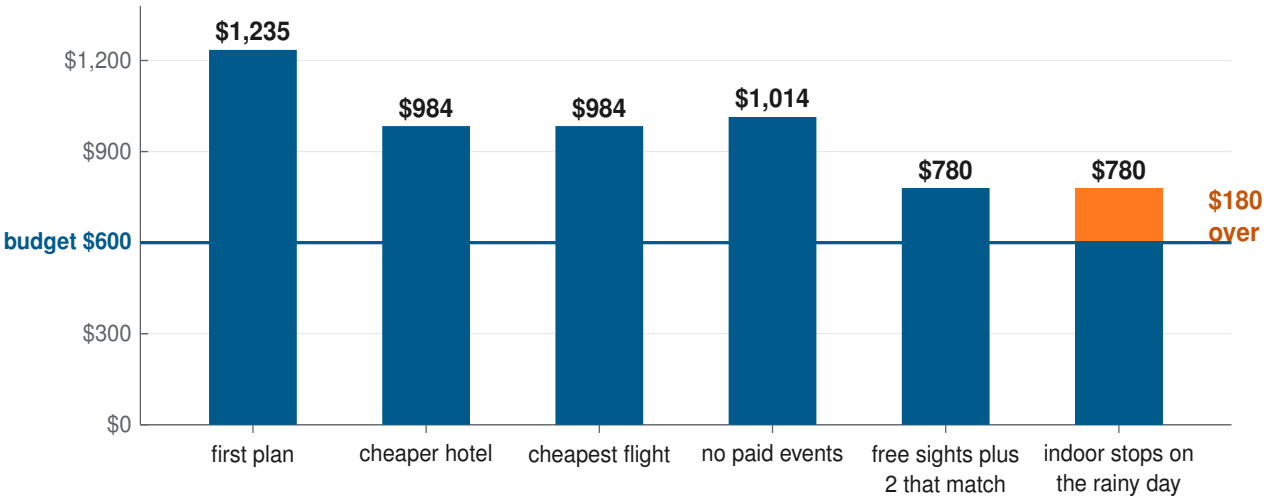
c They turn your plain words into changes



Each request edits the plan instead of starting over.

Why AI agents here? Each step depends on what the last one found, so the AI chooses the next search. Searches the plan cannot do without are run even if the AI skipped them.

Σ Code adds up every price and tries five fixes, from $1,235 to $780



The cost line of Alex's last plan, as the app shows it. Flight and hotel are live prices. Food and local transit are labelled estimates.

flights: **$278**	hotel: **$162**	activities: **$0**	food estimate: **$280**	local transit estimate: **$60**

Not every fix saves money, so code checks the total again after each one.

The AI never adds up the money. Code gives the same exact sum every time, and we can test it.

Still $180 over, Roameo stops and asks Alex what to do

Boston → Chicago · 1 traveler · 3 nights

$780 estimated total

Waiting for your approval

$180 over your $600 budget

The question in the chat, word for word

> The cheapest plan I can build is $780, $180 over your $600 budget. Would you like to raise the budget, shorten to 2 nights, or drive instead of flying?

The plan waits for Alex

✓ Approve plan

Waiting for your approval

a Alex chooses how to close the gap

raise the budget | **2 nights** | **drive**

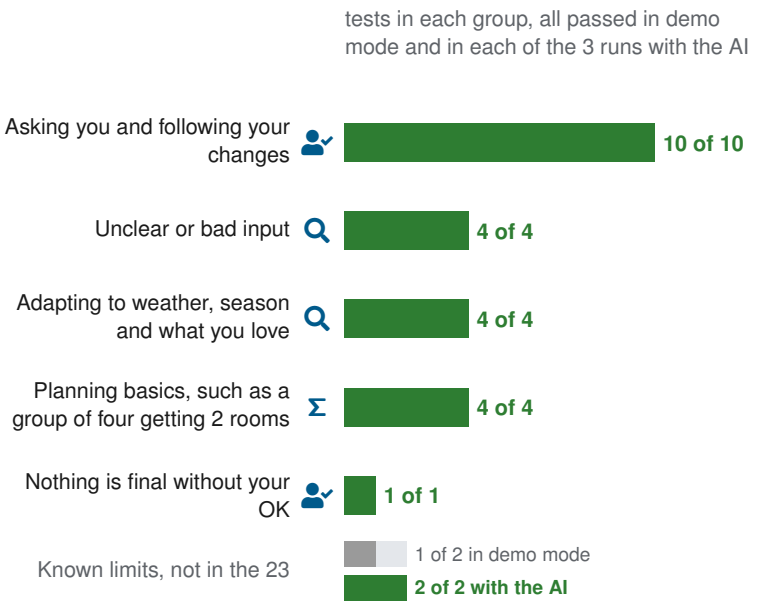
From our tests. With Alex's exact sentence, when the budget is raised, the plan is rebuilt, checked again, and fits. Alex says OK and gets a calendar file.

b Alex can read all 25 steps it took

c Roameo never books or pays
Alex books on a site Alex trusts.

Why it never books. Paying is a decision a person should make, on a site they trust. Saying OK only downloads a calendar file.

All 23 tests of what Roameo should do pass, with and without the AI



What a test is

A conversation we wrote in advance, typed into Roameo, with code that checks the answer. It **passes** when every check comes out right.

Two ways we ran them

Demo mode, practice runs with sample prices, so anyone can try it without sign-ups or paid accounts.

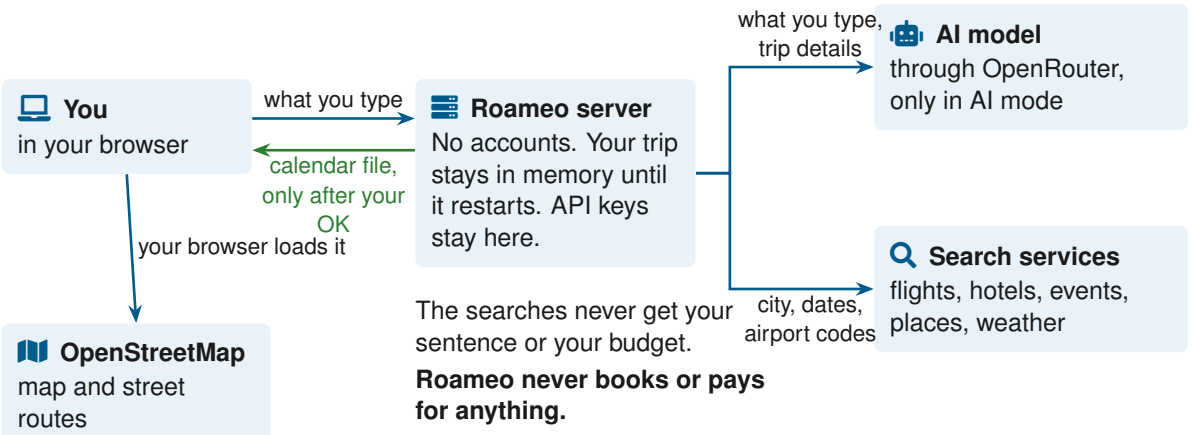
3 runs with the AI, the same tests three times, because the AI can answer differently each time.

- It asks for a missing date.
- On $300 it reports $478 over.
- Nothing counts as approved before OK.

Honest limits. We wrote the tests ourselves, both modes used sample prices and weather, 3 runs is a small sample, and there was no user study. Only the AI understood “first weekend of December”.

Roameo never books or pays, and says what it cannot do yet

What goes where when you use it



Responsible use of AI

It asks before it guesses, and each AI helper has one narrow job.

Σ Code shows its math and labels sample prices and estimates.

You decide, you see every step, and you book it yourself.

What it cannot do yet

- Demo mode covers 8 cities.
- Food and local transit are estimates.
- AI-written summaries and tips can be wrong or out of date.
- No user study yet.

Other apps plan or sell trips, Roameo treats your budget as a limit

Chat assistants
such as ChatGPT and Gemini

Answer questions on any topic, and can write a trip plan in a chat window.

Booking site assistants
such as Expedia's Romie

Helps you plan, shop and book inside Expedia [3].

AI trip planner apps
such as Mindtrip, Layla, Wonderplan

Focus on ideas and day plans, and some add booking, says a review [4].

Also used instead spreadsheets and group chats, a travel agency, and trip organizers such as TripIt that keep bookings you already made.

Where Roameo differs It is not a booking site and not a chat window. It is the planning step in between, where code adds up every price and you decide on the money.

Built for

- students on short trips
- budget travelers
- small groups who split the cost

First group to reach, student clubs. We have not tested the others. This is what each says it is for. [3] Expedia Group, Spring 2024 product release. [4] Fast.io, a review of AI trip planners, 2026.

Anyone can try Roameo free, and a club can share one link

Try it on a laptop

Demo mode starts in about three minutes, with sample prices for 8 cities and no sign-ups or paid accounts.

Share one link with a club

A group of four sees 2 rooms and the cost per person before anyone pays. An optional access code keeps the free searches for the club.

Add live prices one service at a time

About 80 plans a month fit in the free search plan. A class can read the code to see an AI that asks before it spends.

Try it
roameo-l13h.onrender.com

The code is public
github.com/TinyZhen/Travel_Agent

Next, a pilot with 5 to 10 students, to measure overspending and the time to a plan. **Our ask**, try it and tell us where it goes wrong.

AI DID THE LEGWORK, CODE CHECKED THE MONEY, AND ALEX MADE THE CALL

AI agents did the legwork

a Asked Alex where from and when

b Compared 8 flights, 18 hotels and 59 events, and looked indoors for the cold

c Turned plain words like “make it cheaper” into changes

Code checked the money

a Added $278 + $162 + $340 of estimates = $780

b Tried five fixes, from $1,235 down to $780

c Showed the **$180 gap** instead of hiding it

Alex made the call

a Chose how to close the gap

b Could read all 25 steps

c **Raised the budget, the plan fit, Alex said OK** (from our tests)

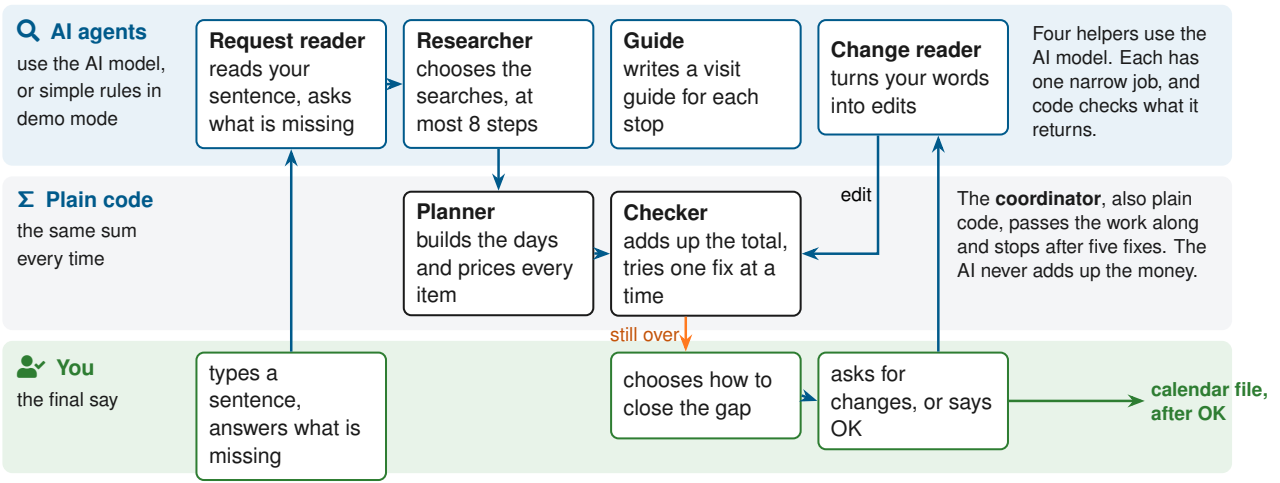
Roameo treats your budget as a limit, not a wish.

A chat reply can tell you a plan fits with nobody checking the sum. In Roameo, AI agents do the searching, plain code adds up every total, and when the money does not work it stops and asks you.

Thank you to the PEAAII AI Applications Hackathon 2026 organizers. Data from Google Flights, Hotels, Events and Maps through SerpApi, Ticketmaster, Open-Meteo and OpenStreetMap. AI model Claude Haiku 5.5 through OpenRouter. Slide template Elegant by Lars Spreng. Logos of the University of Massachusetts Boston, for the authors’ affiliation only.

Scan to try it

Backup, how it works inside



Why six helpers and not one big prompt? Each AI step has a narrow job, and code checks what it returns. Activities come from an allowed list. A trip fact changes only if your own words say so. Searches the plan needs are run even if the AI skipped them, and the researcher stops after 8 steps.

Why not more? A separate helper for every search would add hand-offs without adding decisions.

Backup, the checker's fixes come in a fixed order

The fixed order

1 Indoor stops on a bad weather day, first when one is found

2 A cheaper hotel, one level at a time, which can use two fixes

3 The cheapest flight

4 No paid events

5 Free sights plus the 2 that best match your interests

6 Free sights only

What happened in Alex's run

First plan	$1,235
A cheaper hotel	$984
The cheapest flight	$984
No paid events	$1,014
Free sights plus 2 that match	$780
Indoor stops on the rainy day	$780

The rainy day showed up only after the sights changed, so its fix came last. With five fixes used, Roameo asked Alex.

Budget fixes run only while the plan is over budget. Under the cap of five, the last fix may never run.

Backup, who else helps plan trips, with sources

Who	What it says it is for	How Roameo differs
Chat assistants such as ChatGPT and Gemini	General assistants that answer questions on any topic, travel included.	Code adds up the money, and it asks before it goes over.
Expedia's Romie [3]	An assistant inside Expedia, launched in May 2024 as an early test, for planning, shopping and booking.	Not tied to one booking site, and you choose the trade-offs.
Mindtrip, Layla, Wonderplan [4]	AI trip planners focused on ideas and day plans, some adding booking, says a review.	Built for short trips on a tight budget, with every step shown.
Spreadsheets, group chats, a travel agency, TripIt	Indirect, ways people plan or keep bookings today.	One sentence instead of many tabs, and the total is added by code.

We have not tested the others. This table says what each says it is for, from its own announcement or from a review.

[3] Expedia Group, Spring 2024 product release, expedia.com/newsroom/spring-product-release-2024. [4] Fast.io, Best AI for travel planning 2026, a vendor review and an opinion, fast.io/resources/best-ai-for-travel-planning-2026.

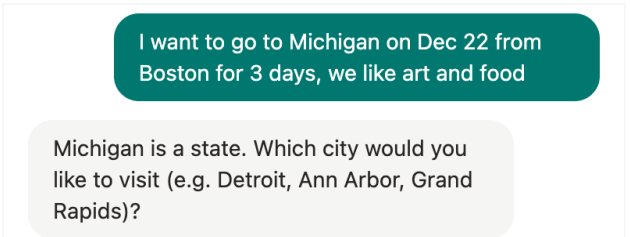
Backup, what the 25 tests check, in plain words

Asking you and following your changes, 10	It asks for a missing date, destination or starting city, and never assumes them. An impossible budget is put to you, and raising it makes the plan fit. “Cheaper” really lowers the price, “different hotel” changes it, a named place is skipped, “more museums” edits the plan and “slower pace” lightens the days.
Unclear or bad input, 4	An unknown city, a past date, vague feedback, and a state instead of a city are all asked about.
Adapting to weather, season and what you love, 4	Rainy days start indoors, winter days keep outdoor time short, a river cruise is not booked out of season, and cutting cost keeps a museum for a museum lover.
Planning basics, 4	A full request gives a priced plan within budget. Driving skips flights and adds gas and parking. A local day trip has no flight or hotel. A group of four gets 2 rooms.
Nothing final without your OK, 1	The plan counts as approved only after you say OK.
Known limits, 2, not in the 23	A city outside the sample data is asked about. A tricky date phrase, where demo mode asks for a date and the AI understands it.

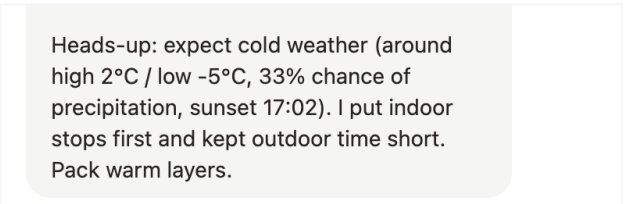
From the results files in the repository, demo mode with 1 run and the AI with 3 runs. The rainy day test passes on sample weather without its rule firing, because sample rain stays at or under 50 percent. The rule did fire in Alex’s live run.

Backup, weather and season change the plan

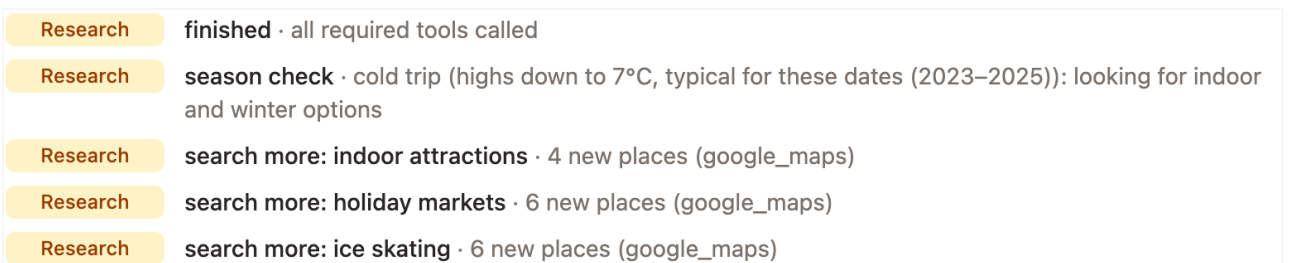
Michigan in December. A state is not a city, so it asks.



Then it warns about the cold and puts indoor stops first.



In Alex's live run, past years' weather for those dates said cold, so the researcher also looked for indoor places, holiday markets and ice skating.



Past years' weather is used when the forecast does not reach that far, and the plan says so.

Visit guides, with the AI on, are written by the AI and labelled so.

Sources

[1] Expedia Group Media Solutions, Path to Purchase research, 2023.
expedia.com/newsroom/eg-path-to-purchase-research

[2] Booking.com, Global AI Sentiment Report, July 2025, through Hospitality Net.
hospitalitynet.org/news/4128268.html

[3] Expedia Group, Spring 2024 product release, which introduced Romie.
expedia.com/newsroom/spring-product-release-2024

[4] Fast.io, Best AI for travel planning 2026, a vendor review and an opinion.
fast.io/resources/best-ai-for-travel-planning-2026

Numbers come from the repository, the evaluation results files, README.md, the code under backend/ and docs/screenshots, or from the sources above.

Thanks

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Data and services Google Flights, Google Hotels, Google Events and Google Maps through SerpApi, Ticketmaster Discovery, Open-Meteo, OpenStreetMap and Leaflet.

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Built with FastAPI, React, Vite and Tailwind CSS. Slides on the Elegant template by Lars Spreng (CC BY 4.0). Poster on the Gemini theme by Anish Athalye (MIT License).

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